

MADEO: Modeler & Analyzer for Direct-wind Electromagnetic Optimization



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Abstract:

The Superconducting Magnet Division (SMD) at Brookhaven National Laboratory (BNL) has the unique capability of developing direct wind magnets, like those that will be used in the Electron-Ion Collider (EIC). This study is the second stage in the development of a new program to model and find optimal configurations for direct wind magnet construction through a new Python based program, complete with a furnished graphical user interface (GUI). Development of a command-line interface version of this program was completed by hand, and then large-language models (LLMs) were used to aid in the development of the GUI and to make the optimization process run much faster than the original version. Through this new program, we were able to effectively find optimal conductor placement for different example magnet configurations. In one example, the program was able to reduce the targeted sextupole harmonic from 782.2 to 0.06, a reduction of about 13,000 times. In the case of the lesser weighted decapole component, it was able to reduce the harmonic from -260.3 to -0.2, a reduction of about 1,300 times in that case. This new program, MADEO, will allow magnet designers at the SMD at BNL to design magnets in a faster and more user-friendly way with better visualizations of the actual magnet designs. This study aligns with Brookhaven's mission to support the Department of Energy's goal of developing higher quality superconducting magnets for future experiments, such as the EIC.

Introduction: Direct-Wind Magnets:

- Direct-wind magnets are superconducting coils wrapped around a support structure, such as an accelerator beam tube (Fig. 1). Because they can carry very high currents, they produce strong magnetic fields, and by changing the coil geometry, those fields can be shaped for different experimental needs.
- In particle accelerators, the strength and type of magnetic field matter because they directly affect how particles move. For dipole magnets, the field strength determines how strongly the beam is bent and therefore links the accelerator's bending radius to the beam energy it can reach (Fig. 2).
- A stronger, cleaner dipole field helps an accelerator reach higher energies with less complexity and energy cost. But dipoles are only one type of accelerator magnet. Quadrupole magnets focus the beam by pushing particles toward the center of the beamline, while sextupole magnets help keep particles with different energies properly focused.



(Fig. 1: A direct-wind magnet being wound at BNL [1])

$$F_{\text{Mag}} = ma_c \Rightarrow qvB = m \frac{v^2}{r} \Rightarrow qvB = (mv) \frac{v}{r} \Rightarrow \boxed{p = qBr}$$
$$E \approx pc \Rightarrow \boxed{E \approx (qBr)c}$$

(Fig. 2: The equation that relates the energy of a particle in a loop to the dipole field applied to it)

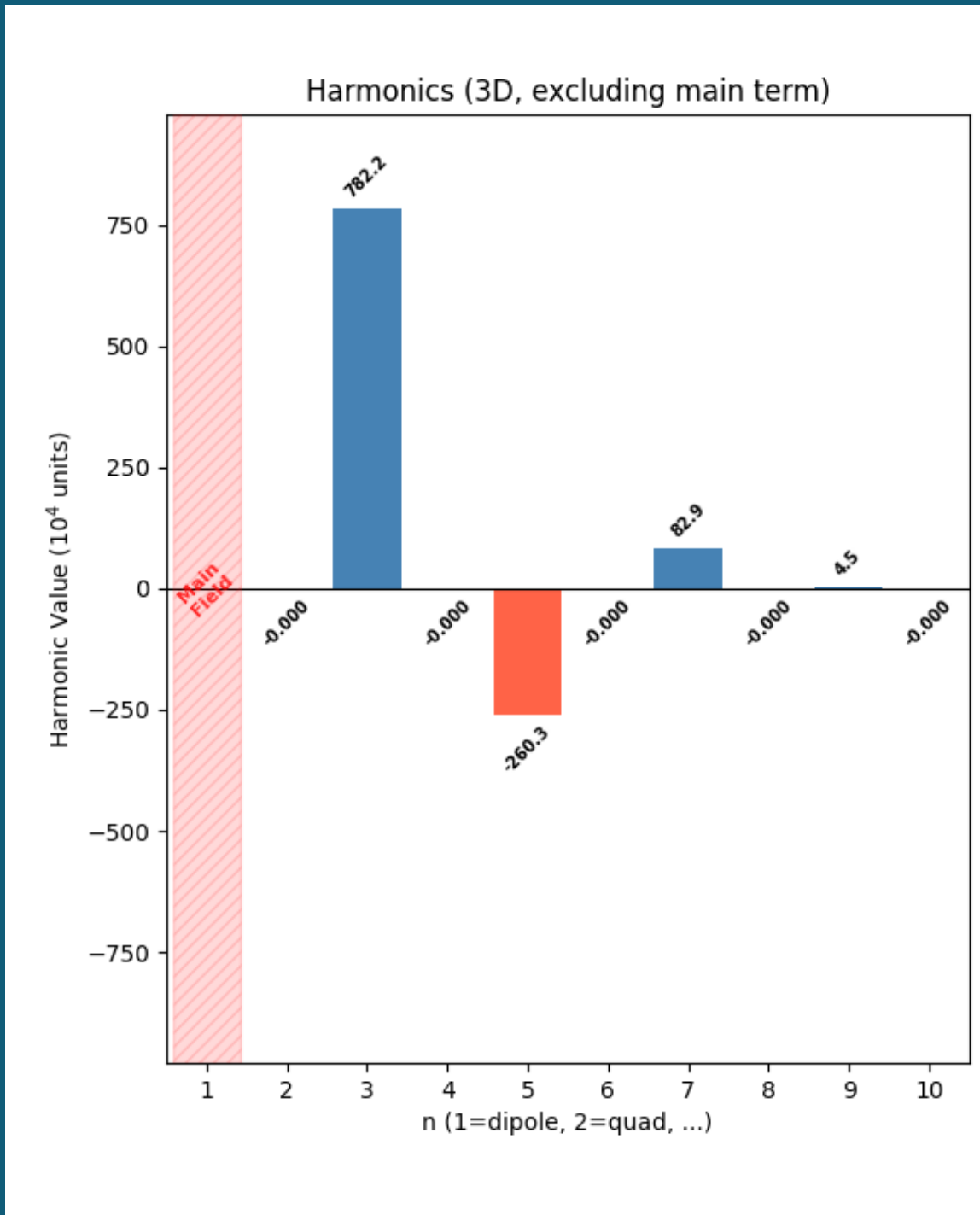
Introduction: Field Harmonics:

- Magnetic fields are composed of an infinite sum of terms called *harmonics* and *skew harmonics*, the mathematical definitions of which are shown in Fig. 3. Each type of magnet, or *pole*, has an associated harmonic term
- To ensure the desired accelerator field, or *main field*, is maximized, we want to minimize undesired field harmonics, so that the main field has the best quality it can have. The region through which we want to minimize these undesired harmonic terms is called the *reference radius*. This is the main radius through which the beam passes in an accelerator
- MADEO is a program that minimizes these undesired harmonics by finding the optimal spacing between conductor blocks

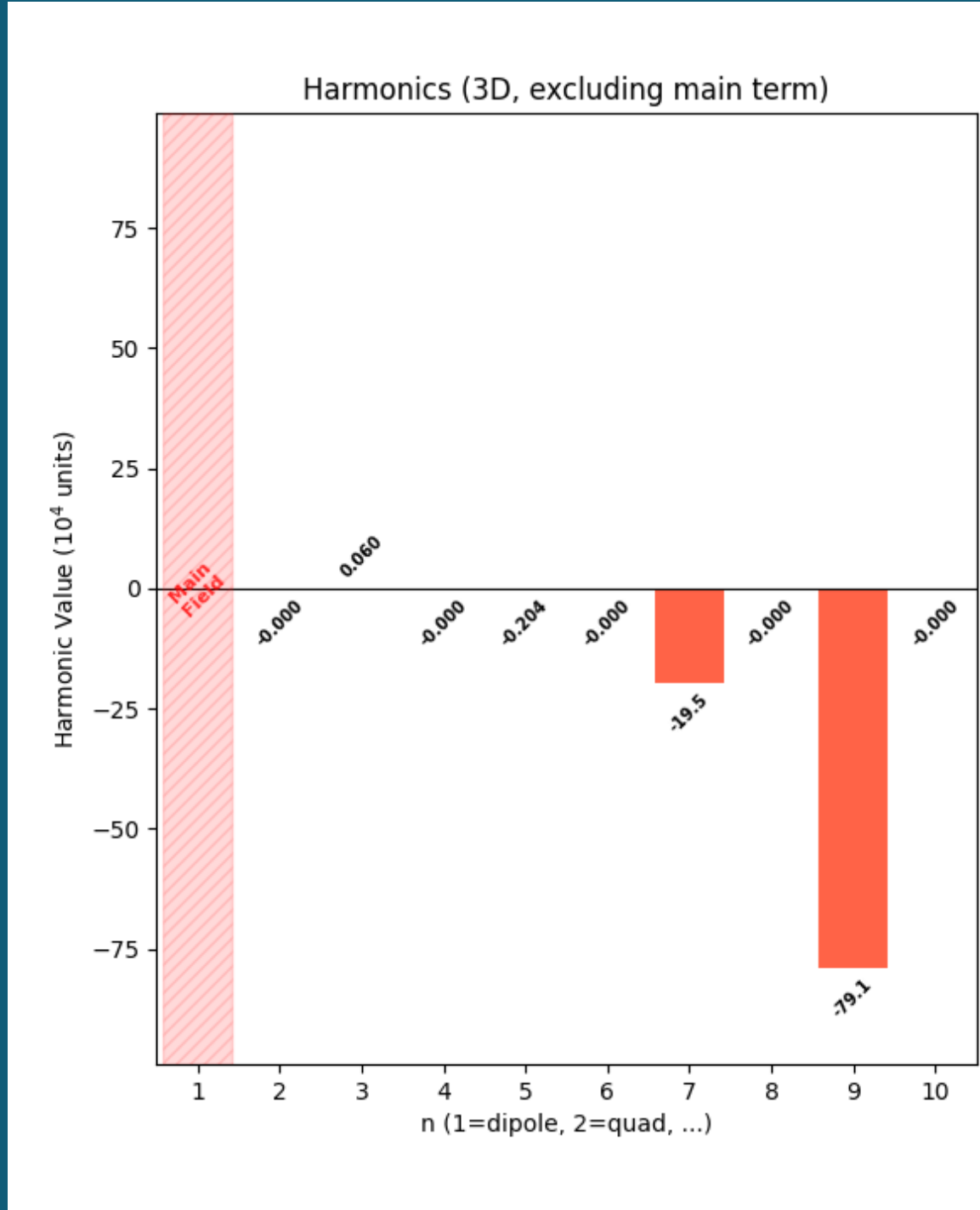
$$b_n = \frac{\mu_0 I}{2\pi} \left(\frac{R_0}{a} \right)^n \cos[(n+1)\phi]$$
$$b_{\text{norm}} = \frac{b_n}{b_{\text{main}}} \cdot 10^4$$
$$a_n = -\frac{\mu_0 I}{2\pi} \left(\frac{R_0}{a} \right)^n \sin[(n+1)\phi]$$
$$a_{\text{norm}} = \frac{a_n}{b_{\text{main}}} \cdot 10^4$$

(Fig. 3: The equations for the harmonics (b) as well as the skew harmonics (a) and their normalized form [2])

Results: Undesired Harmonic Reduction:



(Fig. 6: Harmonics pre-optimization)



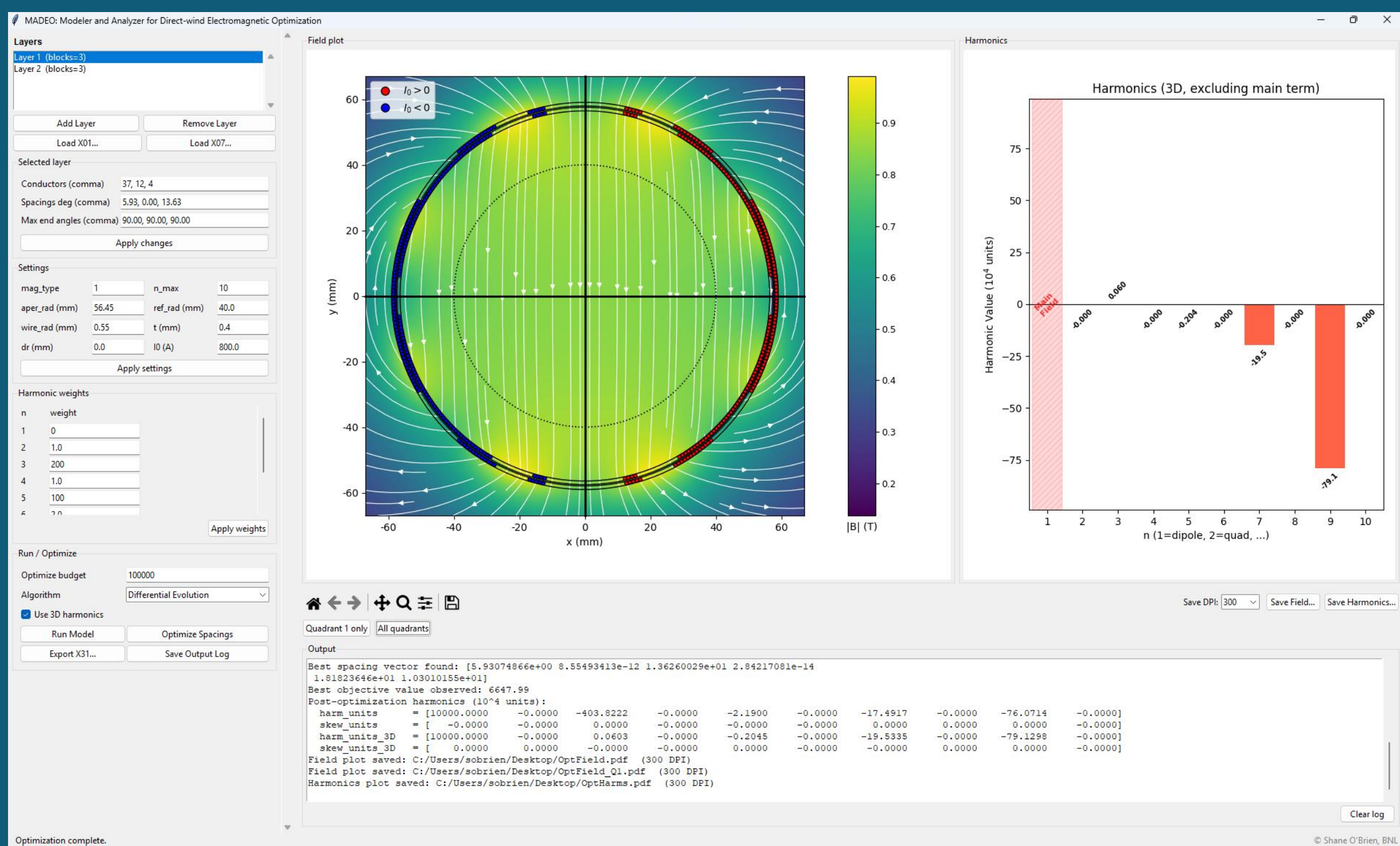
(Fig. 7: Harmonics post-optimization)

Harmonic Component & Weight:	Pre-Optimization:	Post-Optimization:
$n = 1, (w = 0)$	10,000	10,000
$n = 3, (w = 200)$	782.2175	0.0603
$n = 5, (w = 100)$	-260.3244	-0.2045
$n = 7, (w = 1)$	82.8879	-19.5335
$n = 9, (w = 1)$	4.4959	-79.1298

(Table 1: Pre & post-optimization harmonic values)

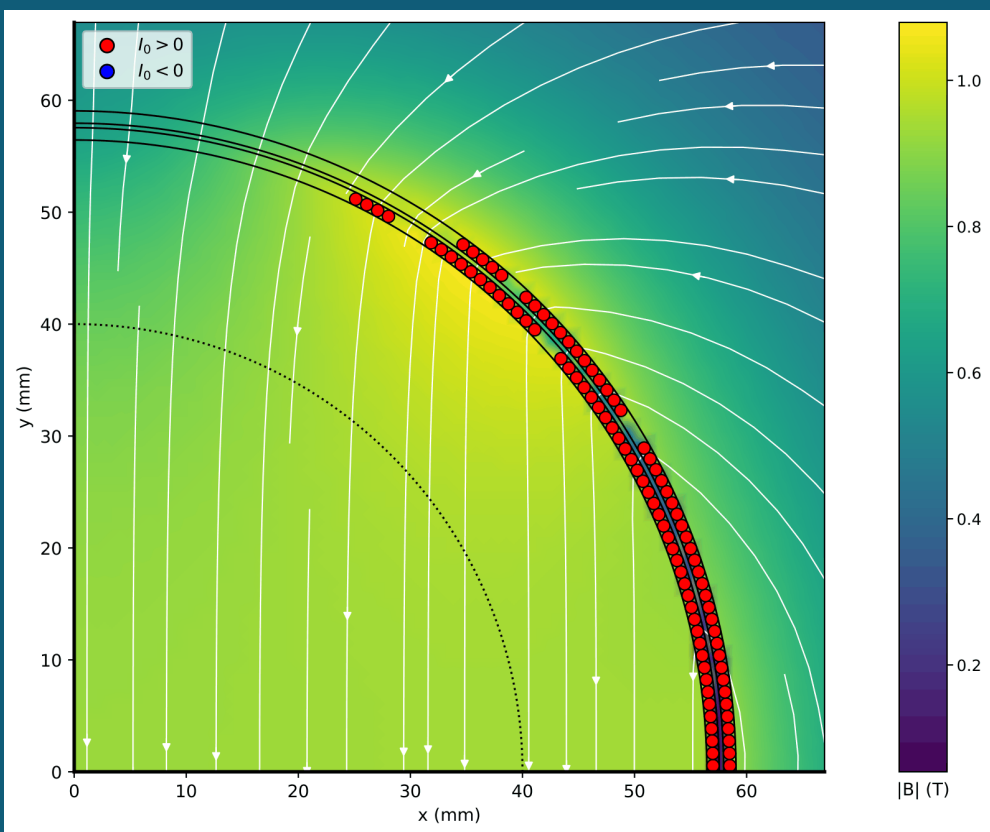
- In this magnet configuration example, the user wanted to mainly suppress the sextupole ($n=3$) harmonic, with a secondary emphasis on suppressing the decapole ($n=5$) harmonic
- We can see that because of this harmonic suppression, the sextupole harmonic was minimized by about 13,000 times its original value and the decapole harmonic, was minimized by about 1,300 times its original value
- Due to the weighting of $n=7$ and $n=9$, these terms are barely even considered by the optimizer

Methods: Modeling & Optimization:



(Fig. 4: The GUI interface for MADEO that a user would see on start-up)

- To use MADEO, the user gives a collection of settings, from which an initial magnet design can be modeled
- These direct-wind magnets are modeled by plotting conductors in an arrangement provided by the user settings. Calculations are then done, of both the individual harmonics and the total magnetic field through the region
- The user can then run this preliminary design through an optimizer, that will then vary the spacing between conductor blocks in each layer until a configuration with an optimal field quality is found, or the given budget is used up
- A differential evolution optimizer is used, as this algorithm gave us very low minimizations on target harmonics in a notably fast time



(Fig. 5: The first quadrant of a dipole magnet modeled in MADEO)

Discussion:

Although the program produced promising results, the final version of the code does not attempt to vary the conductor count within each block. In future work, this capability should be added so that the optimization can further reduce unwanted harmonic terms. In addition, the current version of the code does not account for the iron yoke that is typically placed around the magnet core to help guide the field lines back toward the center of the magnet. Including this feature in a later version would allow the model to better represent the true geometry and field behavior of these magnets. One final addition that would need to be included in a final version as well is considering the insulation of the wire. The wire insulation would impact the harmonics calculated as it would reduce the wire radius.

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NO EXPORT CONTROL

Conclusion:

Through this project, we were able to successfully develop MADEO: a more modern, Python-based program for modeling and optimizing direct wind magnets with an intuitive and easy to use GUI. In addition to developing an algorithm to model and optimize the conductor configurations and the fields they produce, we also laid the groundwork for a more user-friendly replacement to the existing legacy Fortran code currently used to optimize magnet designs. Further adoption of MADEO will allow magnet designers at Brookhaven National Lab to be able to simulate magnets much quicker and not have to worry about editing any sort of files or changing any code themselves. Most importantly, it will give the magnet designers the ability to see their designs and optimization in real time.

References:

- [1] Brookhaven National Laboratory, "Initial Winding of BEPCII Magnets," BNL, <https://www.bnl.gov/magnets/bepcii/initial-winding.php>.
- [2] Gupta, R. *Improving the Design and Analysis of Superconducting Magnets for Particle Accelerators*. Brookhaven National Laboratory, 1996, <https://wpw.bnl.gov/rgupta/wp-content/uploads/sites/9/2023/02/thesisrg.pdf>.